REAR COMBINATION LIGHT

id091800011800

Purpose

- The rear combination lights are used to signal the following conditions to vehicles/people at the rear.
  - — Rear turn lights: Signals a left or right turn of the vehicle.
  - — Brake lights: Signals a vehicle deceleration or stop.
  - — Taillights: Signals the presence of the vehicle to vehicles/people at the rear during nighttime.

Function

- The BCM turns each light on or flashes each light based on each switch signal.

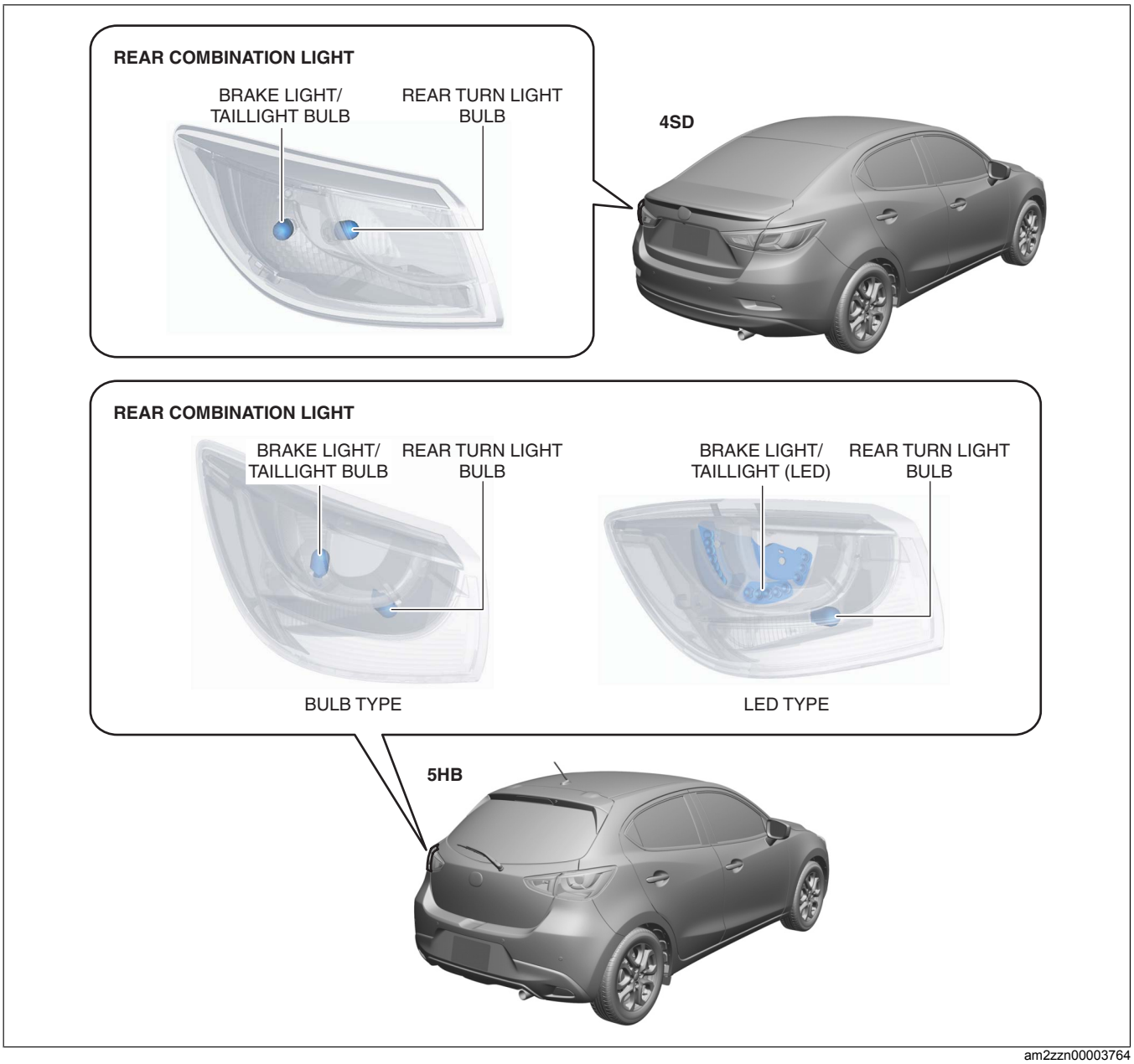
Operation switch	Related light	Operation status
Light switch (TNS)	Taillight	On
Brake switch	Brake light	On
Turn switch	Rear turn light	Flash
Hazard warning switch		

Construction

- The rear combination light groups the following parts:
  - — Rear turn light bulb
  - — Brake light/taillight bulb (rear combination light (bulb type))
- The rear combination light is integrated with the brake light/taillight (LED). (Rear combination light (LED type))
- LEDs have been adopted for the brake light/taillight. (Rear combination light (LED type))

Note

- Fogging or condensation on the inside of the rear combination lights may occur, however, it is a natural phenomenon occurring as a result of a temperature difference between the interior and exterior of the combination lights and it does not affect the light performance.
- Fogging or condensation occurring as a natural phenomenon will dissipate when the temperature inside the rear combination lights rises after the rear turn lights are turned on and a period of time has elapsed.

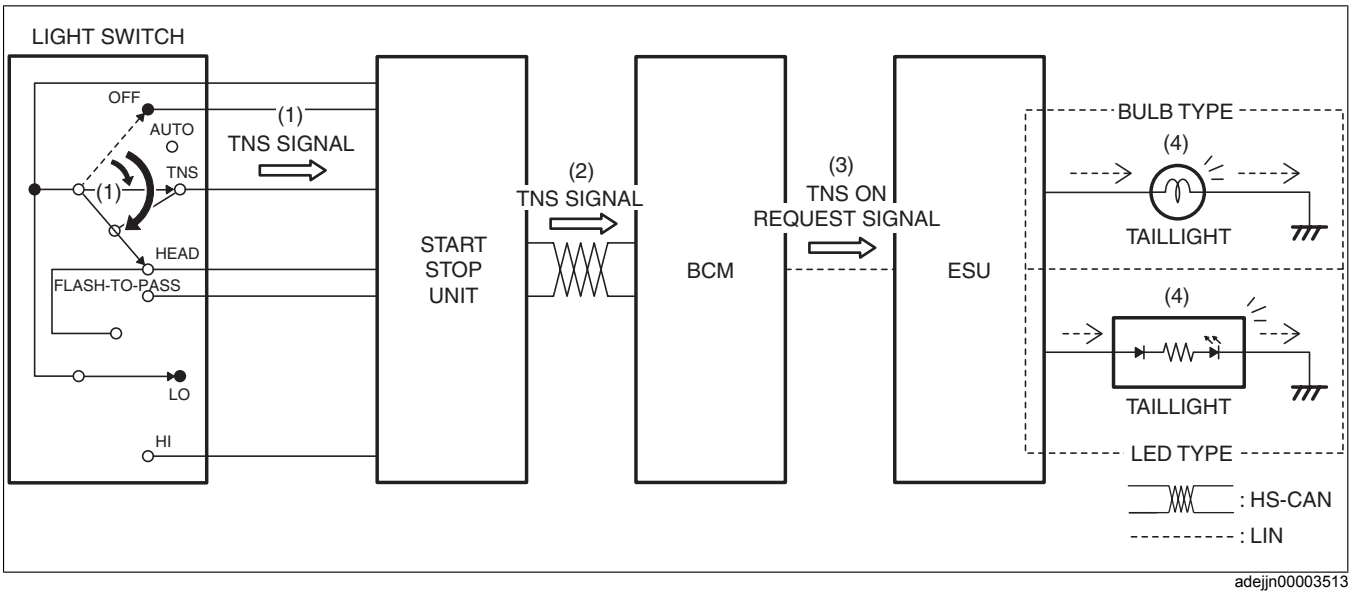


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Operation

Taillight

1. When the light switch is moved to the TNS or HEAD position, the start stop unit detects (1) a TNS signal.
2. The start stop unit sends (2) the TNS signal to the BCM via CAN communication.
3. When the BCM receives the TNS signal, it sends (3) the TNS on request signal to the electrical supply unit (ESU) via LIN communication.
4. When the electrical supply unit (ESU) receives the TNS on request signal, it turns the taillights on (4).



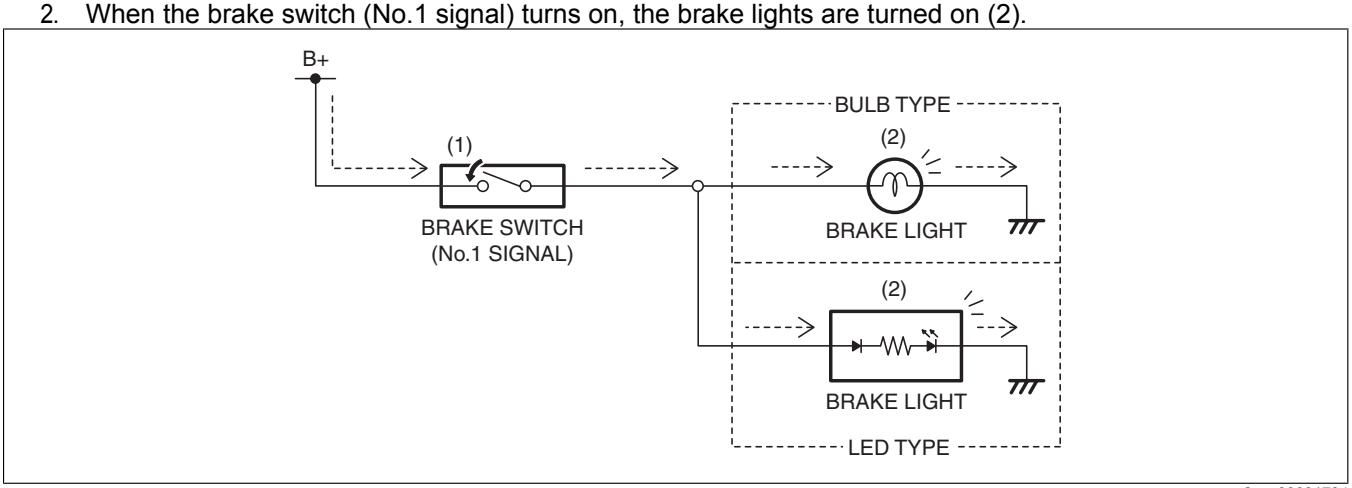
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Brake light

Operation due to brake pedal operation

Vehicles without smart city brake support (SCBS)

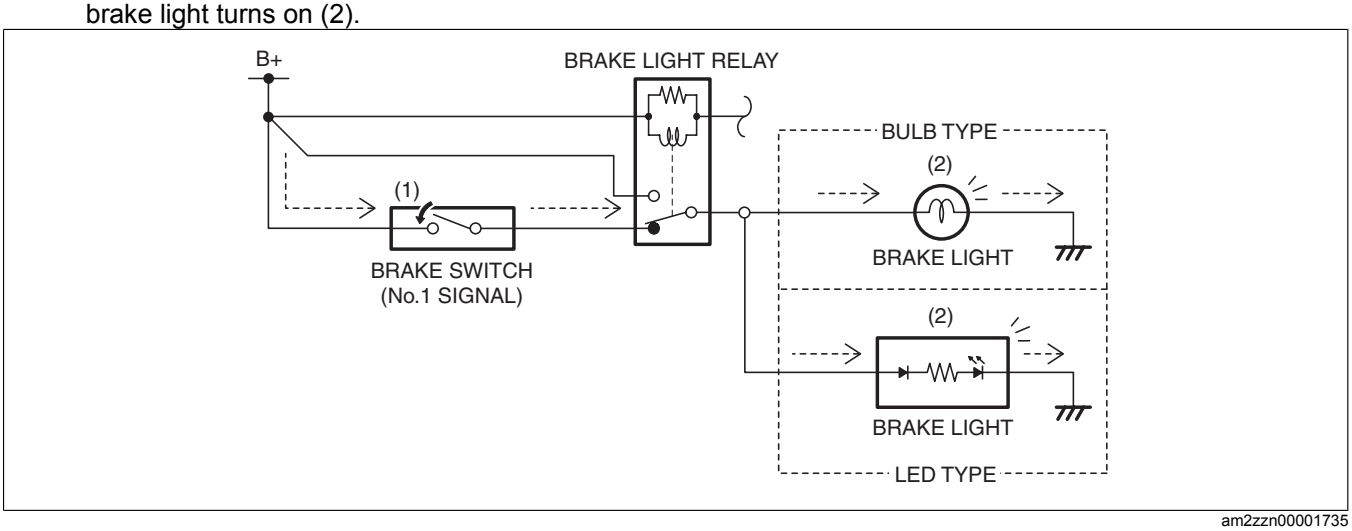
1. When the brake pedal is depressed, the brake switch (No.1 signal) turns on (1).
2. When the brake switch (No.1 signal) turns on, the brake lights are turned on (2).



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Vehicles with smart city brake support (SCBS)

1. When the brake pedal is depressed, the brake switch (No.1 signal) turns on (1).
2. When the brake switch (No.1 signal) turns on while the smart city brake support (SCBS) is not operating, the brake light turns on (2).

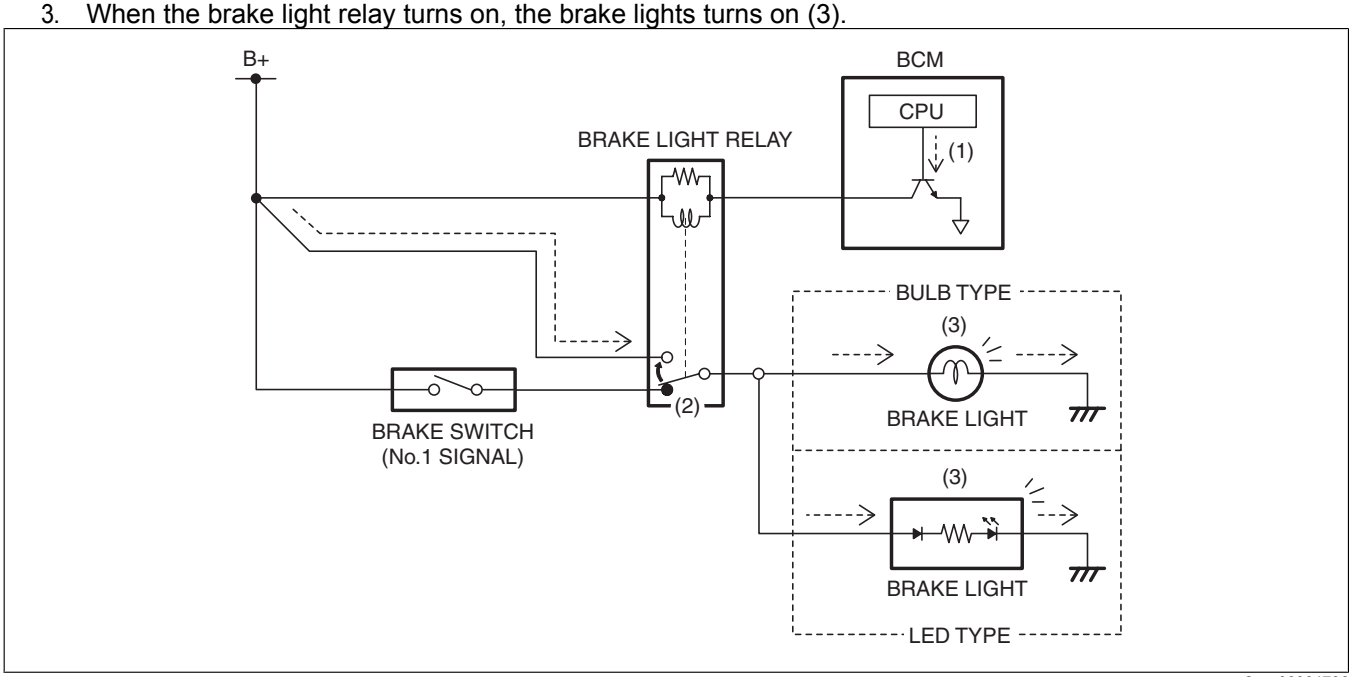


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Operation due to operation request signal from BCM

Note
- For the operation conditions, refer to the [SMART CITY BRAKE SUPPORT (SCBS)]. (See SMART CITY BRAKE SUPPORT (SCBS).)

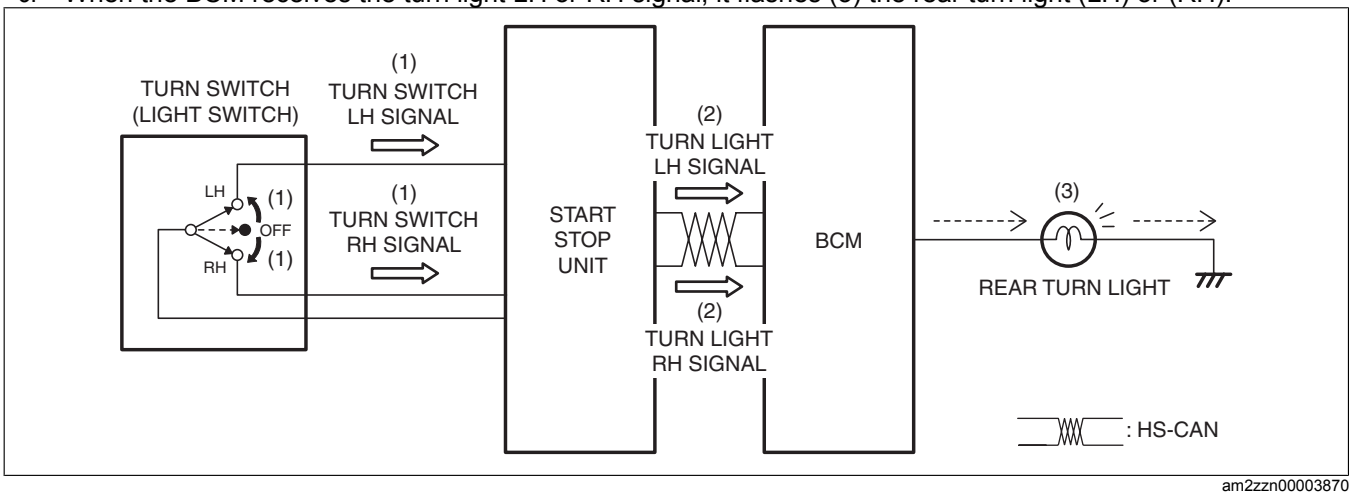
1. When all the operation conditions are met, the BCM turns the transistor on (1) in the BCM.
2. When the transistor in the BCM turns on, the brake light relay turns on (2).
3. When the brake light relay turns on, the brake lights turns on (3).



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Rear turn light
Turn light system

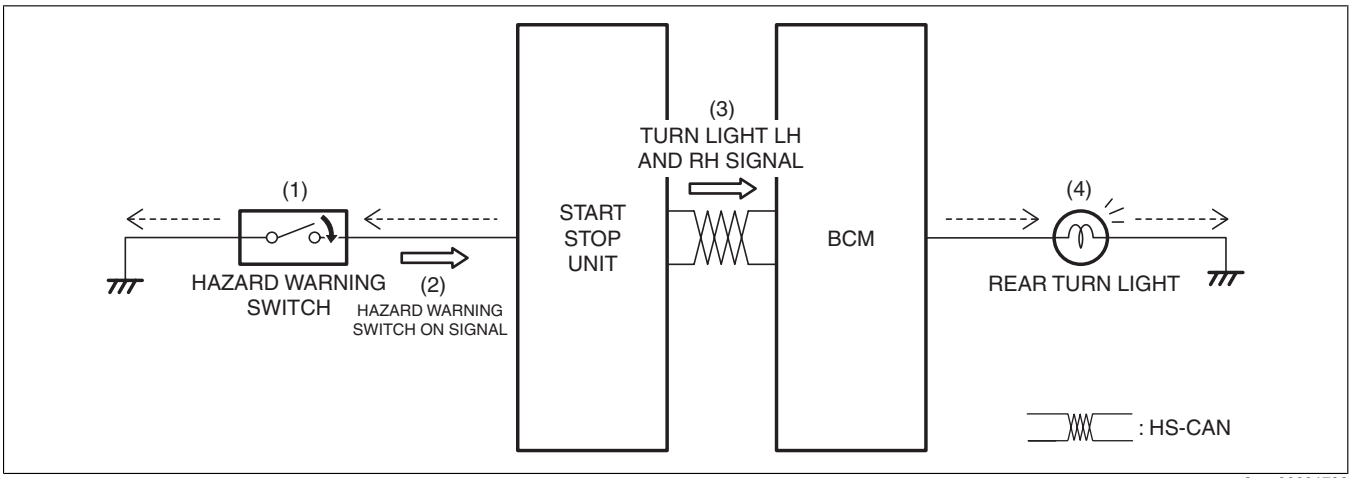
1. When the turn switch is moved to the LH or RH position with the ignition switched ON (engine off or on), the start stop unit detects (1) a turn switch LH or RH signal.
2. The start stop unit sends (2) the turn switch LH or RH signal to the BCM via CAN communication as a turn light LH or RH signal.
3. When the BCM receives the turn light LH or RH signal, it flashes (3) the rear turn light (LH) or (RH).



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Hazard warning system

1. When the hazard warning switch is on (1), the start stop unit detects (2) a hazard warning switch on signal.
2. The start stop unit sends (3) the hazard warning switch on signal to the BCM via CAN communication as a turn light LH and RH signals.
3. When the BCM receives the turn light LH and RH signals, it flashes (4) the rear turn lights (LH) and (RH).



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Fail-safe

- Not applicable